



Premixed flames with dominant diffusion processes stabilized near the source of combustible mixture with the deficient reagent constant concentration

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ABSTRACT

Theoretical analysis performed within the reaction sheet model and numerical simulations undertaken within the model with spatially distributed chemical reactions showed the possibility of the flame existence near the source of combustible mixture with a constant concentration of the deficient reagent. The theory predicts that such flames can exist near the source only in a certain region of combustible mixture velocities which are much lower than the laminar flame velocity. Heat and reagents are transported to the reaction zone by diffusion processes that dominate convective transfer. These properties allow flames with dominant diffusion transfer to be classified in the same phenomena as the flame ball. The results of asymptotic analysis are shown to be in qualitative agreement with the numerical simulations.

1. Introduction

The aim of this work is to describe the features of the still poorly studied, in our opinion, diffusion combustion mode of a premixed combustible gases, which is fundamentally different from a volumetric gas explosion or the propagating waves of chemical reactions - deflagration or detonation. The combustion of premixed combustible gases with dominant diffusion processes includes flame balls in microgravity conditions [1,2], stretched flames at ultra-low gas flow rates [3,4], “weak” flames during gas combustion in microchannels with heated walls [5,6], and other phenomena. A common feature of the mentioned phenomena is that the reagents of the combustible mixture enter the chemical reaction zone due to diffusion and the diffusion fluxes prevail over convective flows. In these cases, combustion occurs in a fairly narrow reaction zone, as in a normal flame. However, if in a normal flame, propagating in a combustible mixture at a certain velocity (flame laminar velocity), the diffusion flux of reagents entering the combustion zone is equal to the convective flux far upstream from the flame zone, then in a diffusion combustion mode, for example, in the case of a stationary flame ball, the reagents enter the reaction zone only due to diffusion. The convective flux of reagents vanishes since the propagation velocity of the reaction front through the combustible mixture is zero in the latter case. The ability of flame stabilization in flows with

characteristic velocities much lower than the laminar flame velocity is a distinctive feature of premixed gas combustion with dominant diffusion processes. The correctness of numerical calculations of flame propagation in combustible gas flows can often be verified by performing estimates of the local flame propagation velocity in a combustible mixture of known composition. For example, if the calculated flame velocity in the simulations turned out to be much lower than the laminar flame propagation velocity, then doubts can arise about the correctness of the numerical scheme or the correctness of the combustion reaction kinetics model. At the same time, the experimentally and theoretically confirmed possibility of combustion wave stabilization in the gas flows with very low velocities raises the problem of identification of these special combustion regimes in order to expand the verification criteria of numerical calculations. Therefore, the description of combustion modes with dominant diffusion processes is an important task that facilitates the developments of correct numerical combustion models in practical devices.

The focus of the paper is the description of diffusion combustion modes of premixed gases near the combustible mixture source. The source peculiarity is that the deficient reactant's concentration of the combustible mixture is sustained constant at the source's surface and it is irrespective to the processes in the source surroundings. Such a surface may be a membrane with selective hydrogen permeability due to

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diffusion, a porous layer separating the combustible mixture and fuel, or the boundary of a supersonic jet through which fast molecules of fuel can escape into the surrounding gas mixture. In the all considered cases, it is assumed that the combustible mixture deficient reactant concentration is sustained constant at the boundary of the region where there are intensive internal nonlinear processes, for example, the process of the deficient reactant generation and consumption leading to establishment of its equilibrium concentration. The constant concentration can sustain by fast convective supply of the reagent at the opposite side of the boundary, which replenishes the deficient reagent concentration much faster than it is consumed in the studied process of slow combustion.

These processes occur so intensively that a decrease or increase in concentration at the boundary is instantly compensated by the action of these processes, maintaining the concentration of the reagent at an equilibrium, constant level. Although the current setup may at first glance appear very specific, a more rigorous consideration of this subject may reveal the sings of such regimes in combustion of mixtures in the regions with slow circulation flows past the flame holders, bluff bodies, channels with abruptly expanding cross section etc. Note that the condition of constant concentration of the combustible mixture deficient component on the surface of the source is similar to the assumption of constant temperature, for example, on a solid wall or on the surface of a porous burner.

The paper considers the problem of flame stabilization near flat and cylindrical surfaces on which the concentration of the deficient component of the combustible mixture is constant. In the models of burner stabilized premixed flames, it is usually assumed absence of the deficient component diffusion inside the porous burner with the combustible mixture flow.

In this case, mixed boundary conditions are used for concentrations and concentration fluxes at the surface of the porous burner [7,8]. In this paper, a special case is considered when the concentration of the deficient reactant is a constant value at a surface. In this case, the flame existence in the gas flow with small gas flow rates is possible, due to diffusion of the reacting components into chemical reaction zone, which prevails over convective transfer. These features allow distinguishing specific premixed gas combustion mode as combustion with dominant diffusion processes.

2. Mathematical model

Fig. 1 shows the schemes of flames stabilized near the planar source (top scheme) and near cylindrical source (bottom scheme) of combustible mixture. The red line at top scheme shows the planar flame front, and the combustible mixture flows with velocity V_0 is highlighted by blue arrows. The bottom scheme shows cylindrical source of combustible mixture with radial gas flow.

The blue domains with large double arrows show the region where internal processes sustain the concentration of combustible mixture at the equilibrium state due to, for example, an intensive turbulent flow.

In the case of planar source of combustible mixture, the equations for the gas temperature T and the concentration of the deficient component C in the localized-reaction-zone model [9] read:

$$T_t + V_0 T_x = \Delta T \quad (1)$$

$$Le (C_t + V_0 C_x) = \Delta C \quad (2)$$

Here $\Delta = \partial^2 / \partial x^2 + \partial^2 / \partial y^2$ is Laplace operator. The localized-reaction-zone model assumes that chemical reactions and heat release occur on the flame surface given by the equation $x = F(y, t)$. The flame surface separates the regions of fresh gas and combustion products, which are indicated by indices 1 and 2, respectively. The equations are written in dimensionless variables, in which the temperature is measured in units of the adiabatic temperature of the laminar flame T_b ,

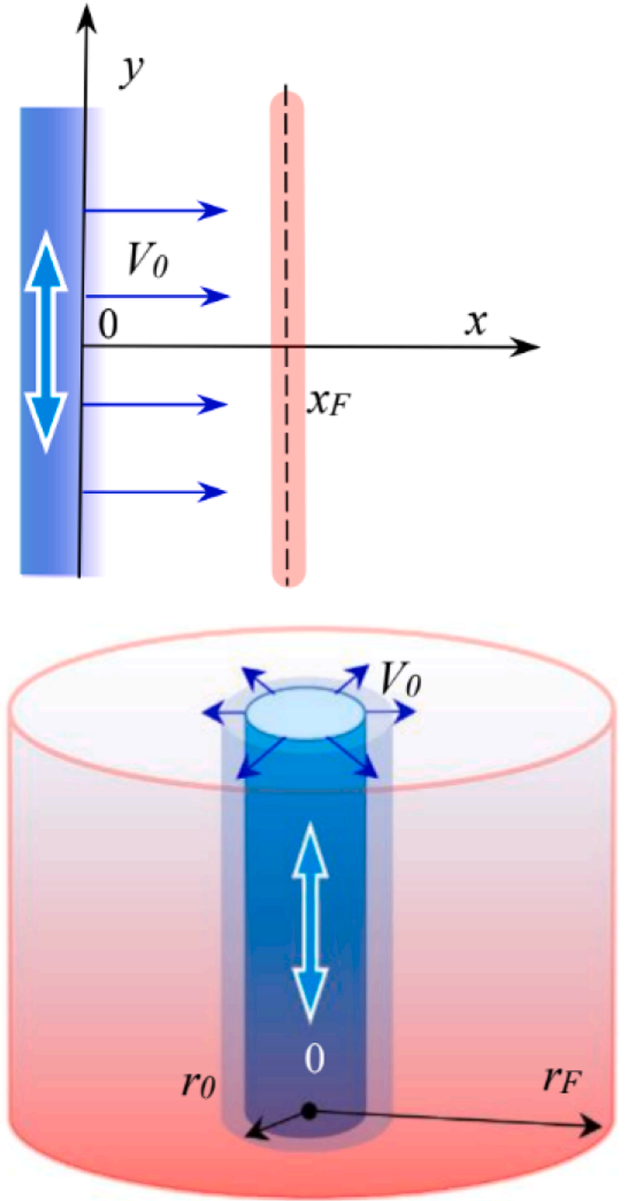


Fig. 1. Schemes of planar and cylindrical flames.

the concentration of the deficient component in units of its concentration C_0 in the fresh mixture. Length is measured in units of laminar flame thermal thickness $l_{th} = D_{th}/U_b$ and time in units of D_{th}/U_b^2 , where D_{th} is the thermal diffusivity of the mixture and U_b is the laminar velocity of the adiabatic flame.

The gas velocity from the burner V_0 is measured in units of U_b . $Le = D_{mol}/D_{th}$ is the Lewis number of the mixture expressed via the molecular diffusion coefficient D_{mol} of the deficient component and D_{th} is the thermal diffusivity of the gas mixture.

The boundary conditions at the flame front $x = F(y, t)$ in the localized-reaction-zone model are [9]:

$$T_1 = T_2 = T_F \quad (3)$$

$$T_{1n} - T_{2n} = (1 - \sigma)U_b(T_F) \quad (4)$$

$$C_{1n} = -Le U_b(T_F) \quad (5)$$

$$C_1 = 0 \quad (6)$$

In the following, we consider the case when the dependence of the flame laminar velocity U_b on the flame temperature T_F has the form:

$$U_b(T_F) = \exp\left[\frac{N}{2}\left(1 - \frac{1}{T_F}\right)\right] \quad (7)$$

Here the following denominations are used:

$\sigma = T_0/T_b$, $= T_a/T_b$, where T_a is activation temperature of the chemical reaction; T_b is the flame adiabatic temperature and T_0 is the initial temperature. The index n in Eqs. (4), (5) denotes the derivative along the normal to the flame front given by the equation $x = x_F(y, t)$. The dependence of the dimensionless normal velocity on the dimensionless temperature in the form of Eq. (7) can be obtained from an asymptotic analysis in which the dimensionless activation energy N is considered as a large parameter. This approximation underlies the localized-reaction-zone model. Note that the dependence of normal velocity on temperature in localized-reaction-zone model can also be constructed using experimental data without addressing to the theoretical background.

The boundary conditions at the surface $x = 0$ in the case of conventional porous burner have the form:

$$T_1 = \sigma \quad (8)$$

$$Le V_0 (C_1 - 1) = C_{1x} \quad (9)$$

These boundary conditions are used before in many papers devoted to burner stabilized flame [7,8,10–12].

In this study we analyzed the special case when the concentration of deficient reactant C_1 is constant at the surface source of the combustible mixture $x = 0$:

$$C_1 = 1 \quad (10)$$

The concentration of the deficient component in the combustion products is zero $C_2 = 0$, and the temperature of the combustion products tends to a constant value in the far field from the flame.

In the case of cylindrical source of combustible mixture, the equations for the gas temperature and the concentration are similar to Eqs. (1) and (2). If the all variables depend on radius r only, the convective term $V_0 T_x$ in Eqs. (1) and (2) are replaced by $V_0 T_x \rightarrow q T_r / r$ and the Laplace operator reads $\Delta = 1/r \partial(r \partial / \partial r)$. Here $q = V_0 r_0$ is the product of the gas velocity V_0 at the source outlet and the radius r_0 of cylindrical source of combustible mixture. The boundary conditions at the flame front $r = r_F$ have the form (3)-(6) in which the derivative along the normal converts into derivative with respect to radius r . The boundary conditions at the surface $r = r_0$ of conventional porous burner have the form:

$$T_1 = \sigma \quad (11)$$

$$q (C_1 - 1) = r_0 C_{1r} \quad (12)$$

In the special case the concentration of deficient reactant at the surface of cylindrical source $r = r_0$ is constant:

$$C_1 = 1 \quad (13)$$

In the next section the stationary solutions in planar and cylindrical geometries are described both for the conventional boundary conditions applied for the burner stabilized flame and for the constant concentration boundary conditions at the source's surfaces. The linear stability analysis of the stationary solutions in the case of planar source of combustible mixture is conducted too, to show differences between solutions describing of flames with dominant diffusion processes and the conventional flame stabilized at the porous burner.

2.1. Stationary solutions in planar configuration of the gas source

Solutions of system (1), (2) with boundary conditions at the source of

the combustible mixture in the form (8), (9), which correspond to the case of a porous burner, are described in many works (see, for example [7,8,10–12]). The relations that follow from these solutions connect the flame front position $x = x_{FS}$ with the combustible mixture flow rate from the burner V_0 and the flame temperature T_F . These relations read:

$$T_F = \sigma + (1 - \sigma)(1 - \exp(-V_0 x_{FS})) \quad (14)$$

$$V_0 = U_b(T_F) \quad (15)$$

According to relations (14), (15) the position of the flame and its temperature are irrespective of the Lewis number. In the case of source with constant concentration, the stationary solutions of Eqs. (1), (2) with boundary conditions (8), (10), have the form:

$$T_{1s} = \sigma + \frac{(T_F - \sigma)(\exp(V_0 x) - 1)}{(\exp(V_0 x_{FS}) - 1)}, \quad T_{2s} = T_F \quad (16)$$

$$C_{1s} = \frac{1 - \exp(Le V_0 (x - x_{FS}))}{(1 - \exp(-Le V_0 x_{FS}))}, \quad C_{2s} = 0 \quad (17)$$

The relations between the flame position x_{FS} , flame temperature T_F and the gas flow rate V_0 assume the form:

$$T_F = \sigma + \frac{(1 - \sigma)(1 - \exp(-V_0 x_{FS}))}{(1 - \exp(-Le V_0 x_{FS}))} \quad (18)$$

$$V_0 = U_b(T_F)(1 - \exp(-Le V_0 x_{FS})) \quad (19)$$

The Eqs. (18), (19) admits solution even in the complete absence of the gas flow $V_0 \rightarrow 0$. In this case the flame stabilizes at the distance x_{F0} from the source surface and it has temperature T_{F0} :

$$T_{F0} = \sigma + \frac{(1 - \sigma)}{Le}, \quad x_{F0} = \frac{1}{Le} U_b^{-1}(T_{F0}) \quad (20)$$

It is interesting that the value of the flame temperature exactly coincides with the flame ball temperature in the absence of the heat losses, and the distance x_{F0} equals to the minimum possible flame ball radius [2, 13]. Fig. 2 shows the dependencies of the flame coordinate x_F on the gas flow rate V_0 , calculated by Eqs. (18), (19) for $\sigma = 0.167$, $N = 11.111$ and $Le = 1.4, 1.2, 0.8$.

The appearance of the $x_{FS}(V_0)$ curves depends on the Le number. If in the case of $Le > 1$ the curves have a U-shaped form and flame

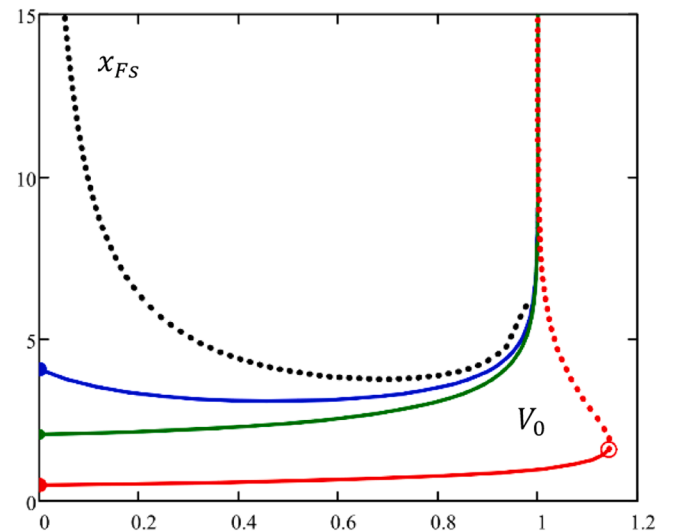


Fig. 2. The flame coordinate x_{FS} depending on the flow velocity V_0 calculated for different Lewis numbers: $Le=1.4$ (blue line), $Le=1.2$ (green line) and $Le=0.8$ (red line). The dashed black curve is the $x_{FS}(V_0)$ dependency in the case of the burner stabilized flame. The dashed red line marks the unstable branch in the case of $Le=0.8$.

stabilization occurs at gas velocities lower than the laminar velocity of a flat flame $V_0 < 1$, then for $Le < 1$ stationary solutions also appear in the region $V_0 > 1$, at gas flow velocities from the burner higher than the normal flame velocity.

Under conditions $Le < 1$, the highly diffusing fuel can migrate preferentially from the source and therefore support higher temperatures, like it occurs in a nonplanar pattern [14]. In the latter case, the curvature of the flame provides excessive enthalpy flow to the flame surface, while in the current setup it is due to the constant concentration of the deficient reagent at the left boundary. Because of the higher effective flame temperature, the flame velocity is larger than would be predicted by a purely one-dimensional analysis for the free propagating flame.

The highest flame velocity is attained at some optimal distance from the fuel source to diminish the heat losses from the flame to the burner. If the distance between flame and burner is large enough the fuel diffusion flux decreases and the flame velocity tends to the flame velocity in the free space.

The black dashed curve at Fig. 2 is the $x_{Fs}(V_0)$ dependency calculated by formulas (14), (15) for the conventional burner stabilized flame. Notice that for conventional burner stabilized flame the flame position is irrespective to the value of Le number.

The solid lines at Fig. 2 show the stable branches of $x_{Fs}(V_0)$ curves. The stability of these branches follows from the linear stability analysis given in Section 3. The linear stability analysis reveals that part of $x_{Fs}(V_0)$ curve is unstable in the case of $Le=0.8$. The unstable part of this curve is shown by dashed red line in Fig. 2. The bifurcation point separating stable and unstable branches is shown by empty red circle. The solid circles at the ends of curves correspond to the flame coordinates x_{F0} in the absence of the gas flow. As it follows from the papers [7,10–12], the left branch of the U-shaped black dashed curve is unstable and the position of bifurcation point depends on the value of Le number.

The dependencies of flame temperature T_F on the flame position x_F are shown in Fig. 3. The curves in Fig. 3 are calculated for the same parameters as for Fig. 2. The color denomination of Fig. 3 curves corresponds to the Fig. 2 curves.

The linear stability analysis of the stationary solutions is described in the Section 3.

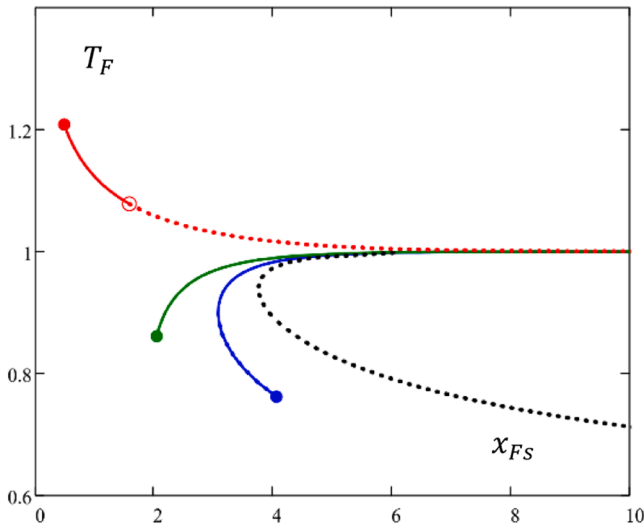


Fig. 3. The flame temperature T_F depending on the flame coordinate x_{Fs} calculated for different Lewis numbers: $Le=1.4$ (blue line), $Le=1.2$ (green line) and $Le=0.8$ (red line). The dashed black curve is the $T_F(x_{Fs})$ dependency in the case of burner stabilized flame. The dashed red line marks the unstable branch in the case of $Le=0.8$.

2.2. Stationary solutions in the case of cylindrical gas source

Let's the radial gas flow is created by cylindrical source with intensity $q = V_0 r_0$, where r_0 is the source radius and the V_0 is the radial gas velocity at the source outlet. The radial velocity in cylindrical geometry is inversely proportional to the radial coordinate. The stationary Eq. (1), (2) describing cylindrical flame can be written in the form

$$(q-1)T_r = rT_{rr} \quad (21)$$

$$(Le q - 1)C_r = rC_{rr} \quad (22)$$

The stationary solutions of Eqs. (21), (22) with mixed boundary conditions (11), (12), have the form:

$$T_1 = \frac{T_F(r_F^q - r_0^q) + \sigma(r_F^q - r_0^q)}{(r_F^q - r_0^q)}, \quad T_2 = T_F \quad (23)$$

$$C_1 = 1 - (r/r_F)^{q Le}, \quad C_2 = 0 \quad (24)$$

In the case of constant concentration at the source surface, described by boundary condition (13), the stationary solution have the form:

$$C_1 = \frac{r_F^{q Le} - r_0^{q Le}}{r_F^{q Le} - r_0^{q Le}}, \quad C_2 = 0 \quad (25)$$

The solution for the temperature is the same as for burner stabilized flame and it is described by formula (23). The algebraic equations related flame temperature T_F and the flame radius r_F in the case of burner stabilized flame read:

$$T_F = \sigma + (1 - \sigma) \left(1 - \left(\frac{r_0}{r_F} \right)^q \right) \quad (26)$$

$$r_F = q U_b(T_F)^{-1} \quad (27)$$

In the case of source with constant concentration the flame temperature and the flame radius can be found from the following equations:

$$T_F = \sigma + (1 - \sigma) \left(\frac{1 - (r_0/r_F)^q}{1 - (r_0/r_F)^{q Le}} \right) \quad (28)$$

$$q = r_F U_b(T_F) (1 - (r_0/r_F)^q) \quad (29)$$

Note that, just as in the case of a flat source geometry, the temperature of a cylindrical flame tends to the adiabatic flame ball temperature T_{F0} in the absence of gas flow $q \rightarrow 0$. Fig. 4 shows the dependencies of the flame coordinate r_F on the gas flow velocity from the source V_0 , calculated by Eqs. (18), (19) for $\sigma = 0.167$, $N = 11.111$ and $Le=1.5, 1.25, 0.75$.

The dependencies in Fig. 4 show that the flame radius increases linearly with input velocity as it becomes sufficiently large, whereas in flat geometry the flame stabilization occurs when the input gas velocity close to the laminar flame velocity.

Solutions for a cylindrical flame at large radii $r_F \gg 1$ transform into solutions for a flat flame, in which the gas velocity V_0 is replaced by the gas velocity $V_F = q/r_F$ at the flame front, and the coordinate x_F is replaced by the flame radius r_F .

3. Linear stability analysis

To study the linear stability of stationary solutions (10)–(11), the variables in the basic equations are written in the form

$$T_{1,2} = T_{1,2s} + \tau_{1,2}\phi, \quad C_1 = C_{1s} + c_1\phi, \quad x_F = x_{Fs} + f\phi \quad (30)$$

All perturbed quantities contain the factor $\phi = \exp(\omega t + iky)$, where ω is the dimensionless growth rate, k is the wave number of perturbations and i is the imaginary unit. The $\tau_{1,2}$, c_1 and f are the amplitudes of

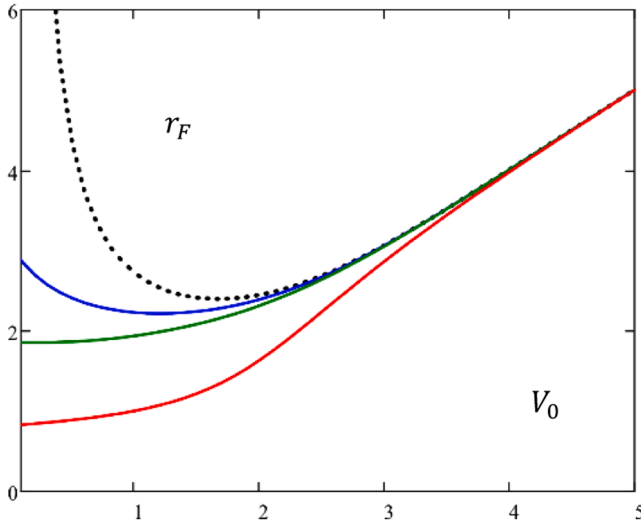


Fig. 4. The flame coordinate r_F depending on the input flow velocity V_0 calculated for different Lewis numbers: $Le=1.5$ (blue line), $Le=1.25$ (green line) and $Le=0.75$ (red line). The dashed black curve is the $r_F(V_0)$ dependency in the case of the burner stabilized flame.

the perturbations, which are assumed to be asymptotically small.

Linear stability analysis was performed in the same way as in paper [8]. We present here an expression for the dispersion equation, which allows us to calculate the dependence ω on k .

$$\begin{aligned} & \frac{Ze^*}{2} \left(\frac{m}{2} - R \text{cth}(R x_{Fs}) \right) \left(\frac{mLe}{2} + R_1 \text{cth}(R_1 x_{Fs}) \right) \\ & + (\omega Le + k^2) \left(\frac{Ze^*}{2} - R(1 + \text{cth}(R x_{Fs})) \right) \\ & = 0 \end{aligned} \quad (31)$$

Here $Ze^* = N(1 - \sigma)/T_{Fs}^2$ is the modified Zel'dovich number, which in the case of free flame is

$Ze = N(1 - \sigma)$. The auxiliary functions R and R_1 in dispersion Eq. (24) are:

$$\begin{aligned} R &= \left(\frac{V_0^2}{4} + k^2 + \omega \right)^{1/2} \quad \text{and} \\ R_1 &= \left(\frac{(V_0 Le)^2}{4} + k^2 + Le \omega \right)^{1/2}. \end{aligned}$$

In the limiting case $x_F \rightarrow \infty$, the dispersion Eq. (24) becomes a dispersion equation for a free flame [13].

Calculations of increment ω by dispersion Eq. (31) with parameters corresponding to flame stabilized near a source with a constant concentration showed that for $Le > 1$ the stationary solutions are completely stable. In this case, all roots of Eq. (31) have a negative real part and the maximum increment is achieved at the wave vector $k = 0$. In the case of $Le < 1$, for example $Le = 0.8$, which corresponds to the red curve in Figs. 2 and 3, the stationary solution is stable, if the flame is located at a smaller distance from the source than the distance corresponding to the turning point of $x_F(V_0)$ curve. The turning point is point of bifurcation. At the bifurcation point, marked by a hollow circle, the real part of the increment is zero and at large coordinate values, the flame becomes unstable. The instability increment has only a real positive part. As the distance between the flame and the source increases, the value of the positive real part of the increment decreases and asymptotically tends to zero. When the flame is removed from the source at a large distance, the flame becomes unstable in a range of wave vectors and the maximum increment is achieved at a certain nonzero wave vector. This case corresponds to the flame diffusion-thermal instability occurring at $Le < 1$ and described in classical works [9,13].

As this type of instability develops, the flame becomes cellular.

4. Simulations within spatially distributed chemical-reaction model

The problem has been also simulated numerically using thermo-diffusive model with spatially distributed single-step chemical reaction (DCR model). Notice, that the localized-reaction-zone model described above may be derived from the DCR model by using common asymptotic analysis [15], assuming a largeness of the activation assuming a largeness of the activation energy $N \gg 1$.

The model contains Eqs. (1) and (2) with additional source terms $(1 - \sigma)W(C, T)$ and

$-W(C, T)$ which were included in the right-hand sides of the Eqs. (1) and (2), correspondingly. Here

$$W(C, T) = \frac{1}{2} N^2 (1 - \sigma)^2 C \exp \left(N \left(1 - \frac{1}{T} \right) \right)$$

is the non-dimensional rate of volumetric chemical reaction. The reaction rate is chosen so that the flame propagation velocity and the temperature of the combustion products in the distributed model are equal to 1 in the dimensionless variables.

The system of the governing equations of DCR model was solved numerically with boundary conditions (8), (10) at the source surface. At the initial moment, the calculated domain is filled with a combustible mixture ($T = \sigma$, $C = 1$), and in the center there is an area of $3l_{th}$ width with a temperature close to adiabatic and zero concentration, which simulates the ignition of the mixture by a heat source. An explicit finite difference scheme of the second order of accuracy in spatial coordinate and the first order of accuracy in time is used in numerical simulation. The size of the numerical grid δ is taken as $\delta = l_{th}/15$, which corresponds to 15 nodes per thermal thickness of the flame l_{th} .

The calculation results for grids with $\delta = l_{th}/20$ are qualitatively the same, and the quantitative difference in the propagation velocity of the adiabatic flame is $< 0.1\%$. Fig. 5 shows the results of calculating the flame coordinate x_{Fs} as a function of inlet velocity V_0 . The flame coordinate corresponds to the position of volumetric chemical reaction maximum. Calculations have shown that in the case of $Le > 1$ there are solutions describing a flame at a large distance from the source, which qualitatively coincides with the analytical calculation within the framework of the localized-reaction-zone model given in Fig. 2. Note that the calculations were performed for a non-stationary model and a stable stationary state was established after a transition period.

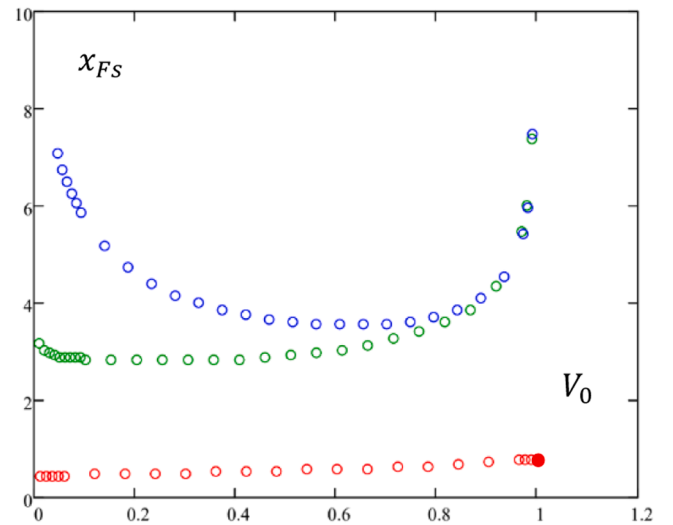


Fig. 5. The flame coordinate x_{Fs} depending on the flow velocity V_0 calculated for different Lewis numbers: $Le=1.5$ (blue circles), $Le=1.2$ (green circles) and $Le=0.8$ (red circles).

In the case of $Le < 1$ stable solution exists only if the distance between the flame and the source is in a limited range of values. This conclusion is consistent with the result of the linear stability analysis, which showed that only part of the curve located below the turning point of $x_{FS}(V_0)$ curve is stable. The limiting maximal flame distance is shown in Fig. 5 by red solid circle.

Fig. 6 shows the concentration, temperature and reaction rate distributions near the source evaluated for $Le = 1.2$ and $V_0 = 0.05$.

The concentration and temperature of the combustible mixture change almost linearly with distance to the source as it follows from Fig. 6. This fact demonstrates a significant difference in the flame structure at low gas flow rates from a freely propagating flame, in which the temperature and concentration distributions have an exponential form.

5. Conclusions

Theoretical analysis performed within the framework localized-reaction-zone model and numerical simulations within the framework of the model with spatially distributed chemical reactions showed the possibility of the flame existence near the source with a constant concentration of the combustible mixture deficient reactant at very low flow rates. The theory predicts that such flames can exist only in a certain region near the source, at velocities of the combustible mixture much less than the laminar flame velocity. This fact shows that such flames differ significantly from propagating flames in free space due to differences in the transport of the heat and the reactants into the reaction zone. In a laminar flame, the diffusion flux of the deficient component is maximal near the reaction zone. The maximum of convective flux of deficient reactant attains far from the reaction zone. The absolute values of the maximums diffusion and convective fluxes are equal.

In the flames described, which can be called diffusion-dominated flames (DDF flame), convective transport of the reactant is much smaller or even absent compared to diffusional transport. Another characteristic feature of these flames is that combustible components enter the reaction zone at speeds smaller than the minimal possible value of the normal speed of nonadiabatic flame [13,16]. These properties of DDF flames allow them to be classified in the same class of phenomena as the flame ball, “weak” flames in heated microchannels and in stretched flow under microgravity conditions. The all these flames were found in experiments after theoretical works which predicted their existence. Note that despite the lack of direct comparison with experiments in which flames described in this paper could be found, proving their existence may be a task for future research. It is possible that such flames can be detected in microgravity conditions, given the history of the flame ball discovery. In any case the presented theoretical results were obtained within the same models as the models describing well known laminar flame, the flame balls and “weak” flames in heated microchannels and stretched flow.

Novelty and significance statement

The novelty of this research is in theoretical description of the premixed gas combustion regime with dominate diffusion processes in planar and cylindrical geometries. These flames can be referred to the same type of premixed gas combustion as a flame ball. It is significant because this study gives new information which facilitates understanding transition from the conventional laminar flame to the experimentally observed flame ball, “weak” flames in heated microchannels and stretched flow in microgravity.

CRediT authorship contribution statement

Sergey Minaev: Investigation, Conceptualization. **Vladimir Gubernov:** Methodology, Investigation. **Evgeniy Sereshchenko:** Software, Investigation.

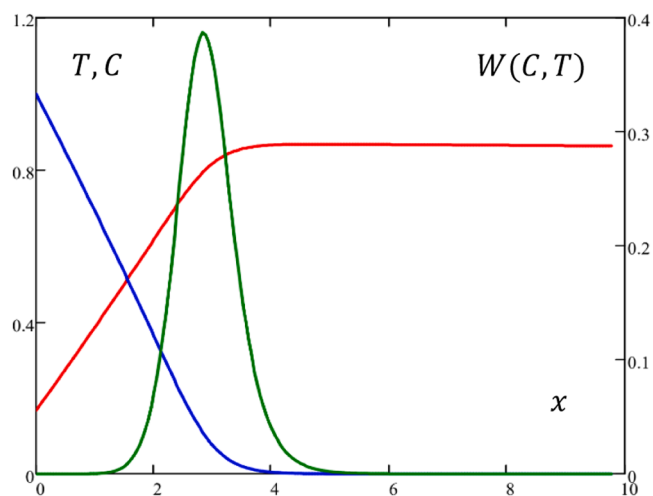


Fig. 6. The concentration (blue line), temperature (red line) and chemical reaction rate (green line) distributions evaluated for $Le = 1.2$, $V_0 = 0.05$, $\sigma = 0.167$ and $N = 11.111$. The right scale refers to nondimensional chemical reaction rate and the left scale refers to nondimensional concentration and temperature.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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